PRANAV SASTRY

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Professional Summary

M.Sc. Computer Science student at TU Darmstadt with hands-on industry experience in Software development, multi-cloud infrastructure, DevOps automation, and AI-powered developer tooling. Experienced with Go, Python, Kubernetes, Jenkins CI/CD, and Terraform. Available for on-site or remote student roles.

Education

Technische Universität Darmstadt - *Master of Science in Computer Science* | Apr 2025 – Ongoing | Darmstadt, DE
Relevant Coursework: Deep Learning, NLP, Statistical Machine Learning, Human Computer Interaction, German

Sri Venkateshwara College of Engineering (VTU) - *Bachelor of Engineering in Computer Science, Artificial Intelligence* | Dec 2020 – May 2024 | Bengaluru, IN
Relevant Coursework: Data Structures, Algorithms, Machine Learning, AI, Computer Networks, Java, Python, DBMS

Work Experience

SAP - *Working Student Software Developer* | Dec 2025 – Present Walldorf, Germany

- Provisioned and configured multi-cloud infrastructure across Azure, AWS, GCP, AliCloud, and SAP's own platform using Terraform, including IAM, networking, and backend configurations from scratch.
- Designed and published an AI skill extension using Claude Code that automates cloud environment onboarding across 6 cloud providers - handling file generation, prerequisite checks, branch creation, and PR opening; adopted by developers across teams.
- Extended Jenkins-based CI/CD pipelines to support a new cloud provider with secure credential handling, deployment configurations, and environment-specific parameters for fully automated deployments.
- Developed backend Go modules to automate secrets management - reading from a secrets store, writing to target environments with optional password generation, structured serialization, and manual verification steps.
- Maintained internal technical documentation, service registries, monitoring dashboards, and onboarding guides aligned with deployed infrastructure state.

Freelance Web Developer - *Freelance* | Jul 2024 – Oct 2024 | Bengaluru, India

- Developed and maintained an e-commerce website with focus on usability and backend data automation using Python.
- Built Python scripts to manage product data and automate update workflows, reducing manual effort by over 40% in testcases and services.

Nokia Solutions and Networks - *DevOps Intern* | Aug 2023 – Jun 2024 | Bengaluru, India

- Automated CNF/VNF testing and certificate validation in Kubernetes environments using Python, Shell, and YAML, reducing manual effort by 60%.
- Built Docker-based failover and health check scripts for Kubernetes pods, improving system reliability by 40%.
- Deployed Grafana/JMeter dashboards for real-time performance monitoring and auto-generated structured reports (Excel/HTML) with Python.
- Developed scripts using Python, YAML, and Shell to automate VM setup including package installations, IP configurations, and software deployment.

- Automated security handshakes and certificate management using Shell scripting, ensuring secure communication protocols.

Hindustan Aeronautics Limited (HAL) - *Data Analyst Intern* | Aug 2023 – Sep 2023 | Bengaluru, India
- Analyzed helicopter Maintenance, Repair, and Overhaul (MRO) data using Python (Pandas, NumPy) and Excel at the Rotary Wing Research and Design Centre.
- Supported predictive maintenance initiatives by analyzing failure rates and optimizing maintenance schedules.

Bambhari.com - *Machine Learning Intern* | Jun 2023 – Aug 2023 | Bengaluru, India
- Led a team to develop a CNN-based disease detection model using OpenCV for analyzing urine samples based on the Thaila Bindu Pariksha method.
- Managed task distribution, cross-functional coordination, and real-time ML prototype deployment.

Projects

E-Waste Classification & Sorting System | TensorFlow, OpenCV, MobileNet SSD-V2 | 2024
- Developed a TensorFlow MobileNet SSD-V2 model with an OpenCV pipeline for live e-waste recognition, achieving 90% accuracy.
- Handled data annotation, preprocessing, and model optimization; research published as a journal paper (IEEE).

Online Auctioning System | Django, Python, REST APIs | 2023 - GitHub
- Built a full-stack Django web application with real-time auction features, admin dashboard, secure authentication, and real-time bidding.

Technical Skills

Programming & Scripting: Python, Go, C, JavaScript, NodeJS, SQL, YAML, Bash, HTML, CSS
Cloud & Platforms: Azure, GCP, SAP BTP, SAP CCloud
DevOps & Infrastructure: Terraform, Kubernetes, Docker, Jenkins, GitOps, GitHub Actions, GitLab CI/CD, Linux
Monitoring & Security: Grafana, JMeter, InfluxDB, OpenSSL, HashiCorp Vault
AI & Tooling: Claude Code, GitHub Copilot, MCP (Model Context Protocol), AI skill/extension development, Prompt Engineering
Frameworks & Libraries: TensorFlow, OpenCV, Pandas, NumPy, Scikit-learn, Django, REST APIs
Collaboration: Jira, Agile/Scrum, Git

Certifications

- Certified Kubernetes Administrator (CKA)
- Microsoft Certified: Azure Fundamentals
- Positioning SAP Business Data Cloud
- Google Cloud Program (GCP)
- GitHub Copilot
- IELTS 7.0 (C1 English)

Publication

A Deep Learning Approach for Electronic Waste Management and Classification in Smart Cities, IEEE-2026

Languages

English (C1, Full Professional) | German (A2, Limited Working) | Telugu (Native) | Kannada (Native) | Hindi (Full Professional)